

Percentage

Percent is a value when checked w.r.t. 100.

like, If you get 80% on a test,
you got 80 out of 100 questions right.

⇒ Three Comparison tools →

- 1) Difference
- 2) Ratio
- 3) Percentage

examples →

› A person spends 20% of his salary.

20% ⇒ 20 out
of 100

If salary is 100, expense is 20.

$$20\% = \frac{20}{100} = \frac{1}{5} \quad \left\{ 20\% = \frac{1}{5} \right\}$$

› If he spends 25% of his salary.

$$25\% = \frac{25}{100} = \frac{1}{4} \quad \left\{ 25\% = \frac{1}{4} \right\}$$

It means, if a whole part is 100,
we are talking about 25 out of 100
or we say, one-fourth.

Ratio $\rightarrow$ Ratio is a comparison by division

Boy 1 have 20 chocolates,

Boy 2 have 30 chocolates, find the ratio.

Boy 1 : Boy 2

20 : 30

2 : 3

{ Ratio should
be solved
in its
lowest terms. }

Or, can be written as,

$$\frac{\text{Boy 1}}{\text{Boy 2}} = \frac{20}{30} = \frac{2}{3}$$

Percentage is also a ratio compared to 100.

Boy : Girl

40 : 60

4 : 6

2 : 3

$$\frac{\text{Boy}}{\text{Girl}} = \frac{40}{60} = \frac{4}{6} = \frac{2}{3}$$

Or

Boys in class = 40

Girls in class = 60

Now,

$\Rightarrow$ Boys contribute to what portion of class?

Whole class = $(2+3) = 5$

Boys = 2

Ratio = $\frac{2}{5} \rightarrow$ Part of Boys
 $\rightarrow$ whole class

$\Rightarrow$ Girls contribute to what portion of class?

Ratio = $\frac{3}{5}$ (Girls)
(whole class)

⇒ Amit obtained 30 marks out of 75 marks, find his marks percentage.

$$75 \longrightarrow 30$$

$$1 \longrightarrow \frac{30}{75}$$

$$100 \longrightarrow \left(\frac{30}{75} \times 100 \right)$$

Here we are taking total marks as 100.

{ Percentage is always computed out of 100 }

$$\frac{30}{75} \times 100 = 40\%$$

⇒ Total Salary →	Chintu	Mintu
Expenses →	1500	1800
	225	270

• Chintu

$$\text{Percentage} \Rightarrow \frac{225}{1500} \times 100 = 15\%$$

• Mintu

$$\text{Percentage} = \frac{270}{1800} \times 100 = 15\%$$

⇒ Exercise →

$$15\% = \frac{15}{100} = \frac{3}{20}$$

$$75\% = \frac{75}{100} = \frac{3}{4}$$

$$25\% = \frac{25}{100} = \frac{1}{4}$$

$$35\% = \frac{35}{100} = \frac{7}{20}$$

$$80\% = \frac{80}{100} = \frac{4}{5}$$

$$45\% = \frac{45}{100} = \frac{9}{20}$$

$$\cdot \frac{8}{40} \Rightarrow \frac{8}{40} \times 100 = 20\%$$

$$\cdot \frac{5}{25} \Rightarrow \frac{5}{25} \times 100 = 20\%$$

$$\cdot \frac{8}{50} \Rightarrow \frac{8}{50} \times 100 = 16\%$$

$$\cdot \frac{9}{45} \Rightarrow \frac{9}{45} \times 100 = 20\%$$

$$\cdot \frac{16}{80} \Rightarrow \frac{16}{80} \times 100 = 25\%$$

$$\cdot \frac{4}{10} \Rightarrow \frac{4}{10} \times 100 = 40\%$$

$$\begin{aligned} \cdot 24\% &= (4\%) \times 6 \\ &= \frac{1}{25} \times 6 \\ &= \frac{6}{25} \end{aligned}$$

(or)

$$24\% = 24/100$$

$$24\% = \frac{24}{100} = \frac{6}{25}$$

$$24\% = \frac{6}{25} \rightarrow \text{25 is Base value}$$

$\xleftarrow{ka}$

$$\begin{aligned} \cdot 37.5\% &= (12.5\%) \times 3 \\ &= \frac{1}{8} \times 3 \end{aligned}$$

$$37.5\% = \frac{3}{8}$$

$\xleftarrow{ka}$

$$\begin{aligned} \cdot 62.5\% &\Rightarrow 12.5\% \times 5 \\ &= \frac{1}{8} \times 5 \end{aligned}$$

$$62.5\% = \frac{5}{8} \rightarrow \text{Base value}$$

$\xleftarrow{ka}$

$$\begin{aligned} \cdot 48\% &= 4\% \times 12 \\ &= \frac{1}{25} \times 12 \end{aligned}$$

$$48\% = \frac{12}{25}$$

$$\begin{aligned} \cdot 22.22\% &\Rightarrow (11.11\%) \times 2 \\ &= \frac{1}{9} \times 2 \end{aligned}$$

$$22.22\% = \frac{2}{9}$$

$$\begin{aligned} \cdot 60\% &= 20\% \times 3 \\ &= \frac{1}{5} \times 3 \end{aligned}$$

$$60\% = \frac{3}{5}$$

$$\begin{aligned} \cdot 36.36\% &\Rightarrow 9.09\% \times 4 \\ &= \frac{1}{11} \times 4 \end{aligned}$$

$$36.36\% = \frac{4}{11}$$

$$\begin{aligned}
 \bullet \quad 325\% &\Rightarrow 300\% + 25\% \\
 &= 3 + \frac{1}{4} \\
 &= \frac{13}{4}
 \end{aligned}$$

$$\begin{aligned}
 \bullet \quad 416.66\% &\Rightarrow 400\% + 16.66\% \\
 &= 4 + \frac{1}{6} \\
 &= \frac{25}{6}
 \end{aligned}$$

$$\begin{aligned}
 \bullet \quad 152\% &\Rightarrow 100\% + 52\% \\
 &= 1 + \frac{13}{25} \\
 &= \frac{38}{25}
 \end{aligned}$$

$$\begin{aligned}
 \bullet \quad 180\% &\Rightarrow 100\% + 80\% \\
 &= 1 + \frac{4}{5} \\
 &= \frac{9}{5}
 \end{aligned}$$

$$\begin{aligned}
 \bullet \quad 502\% &\Rightarrow 500\% + 2\% \\
 &= 5 + \frac{1}{50} \\
 &= \frac{251}{50}
 \end{aligned}$$

$$\begin{aligned}
 \bullet \quad 610\% &\Rightarrow 600\% + 10\% \\
 &= 6 + \frac{1}{10} \\
 &= \frac{61}{10}
 \end{aligned}$$

$$\begin{aligned}
 \bullet \quad 412.5\% &\Rightarrow 400\% + 12.5\% \\
 &= 4 + \frac{1}{8} \\
 &= \frac{33}{8}
 \end{aligned}$$

$$\begin{aligned}
 \bullet \quad 260\% &\Rightarrow 200\% + 60\% \\
 &= 2 + \frac{3}{5} \\
 &= \frac{13}{5}
 \end{aligned}$$

$$\begin{aligned}
 \bullet \quad 277.77\% &\Rightarrow 200\% + 77.77\% \\
 &= 2 + \frac{7}{9} \\
 &= \frac{25}{9}
 \end{aligned}$$

$$\begin{aligned}
 \bullet \quad 225\% &\Rightarrow 25\% \times 9 \\
 &= \frac{1}{4} \times 9 \\
 &= \frac{9}{4}
 \end{aligned}$$

Percentage Increase/Decrease →

(I) If a number is increased by 20%.

Let Number = 100
Increase = + 20
New = 120

Initial : New
100 : 120
5 : 6

$$20\% = \frac{1}{5}$$

Initial : New
5 → 6
+1
5 : 6

(II) If a Number is increased by 50%.

Let Number = 100
Increase = + 50
New = 150

Initial : New
100 : 150
~~10~~ : ~~15~~
2 : 3

$$50\% = \frac{1}{2}$$

Initial : New
2 → 3
+1
2 : 3

(III) If a Number is increased by 37.5%.

Let Number = 100
Increase = + 37.5
New = 137.5

Initial : New
100 : 137.5
8 : 11

$$37.5\% = \frac{3}{8}$$

ka

Initial : New
8 → 11
+3
8 : 11

(IV) If a Number is increased by 36.36%.

Let Number = 100

Increase = +36.36

New = 136.36

$36.36\% = \frac{4}{11}$

Initial : New

100 : 136.36

1 : $\frac{15}{11}$

11 : 15

Initial : New

11 : 15
+4

11 : 15

⇒ What is 24% of 2400?

$\frac{24}{100} \times 2400 = 24 \times 24$
 $= 576$

⇒ What is 16.67% of 1680?

$\frac{1}{6} \times 1680 = 280$

$\left\{ 16.67\% = \frac{1}{6} \right\}$

⇒ 24 is what percent of 80?

$\frac{24}{80} \times 100 = \frac{24^3}{8} \times 10$
 $= 30\%$

⇒ 29 is what percent of 232?

$\frac{29}{232} \times 100 = \frac{100}{8}$
 $= 12.5\%$